Problem Set 5, COSC330/PHYS306
Chapters 6–9

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1 Chapter 6

1. **[Section 6-4, Exercise 13]**
   The waveforms in Figure 6–73 are applied to the comparator as shown. Determine the $(A = B)$ waveform.

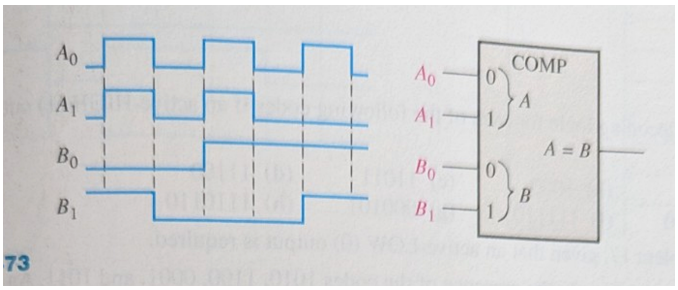


FIGURE 6–73

2. **[Section 6-4, Exercise 14]**
   For the 4-bit comparator in Figure 6–74, plot each output waveform for the inputs shown. The outputs are active-HIGH.

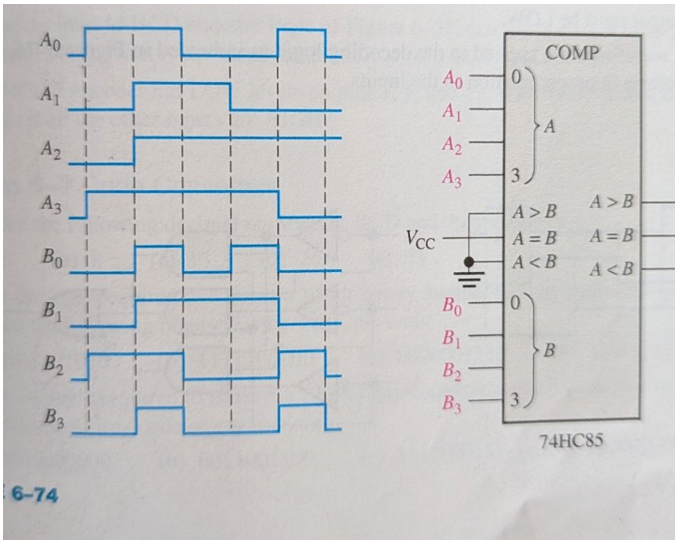


FIGURE 6–74

3. **[Section 6-5, Exercise 20]**
   If the input waveforms are applied to the decoding logic as indicated in Figure 6–76, sketch the output waveform in proper relation to the inputs.

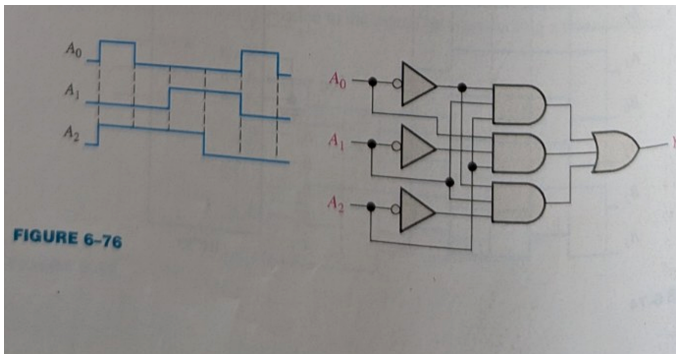


FIGURE 6–76

4. **[Section 6-8, Exercise 28]**
   For the multiplexer in Figure 6–79, determine the output for the following input states:

$$D_0 = 0, \quad D_1 = 1, \quad D_2 = 1, \quad D_3 = 0,$$

$$S_0 = 1, \quad S_1 = 0.$$

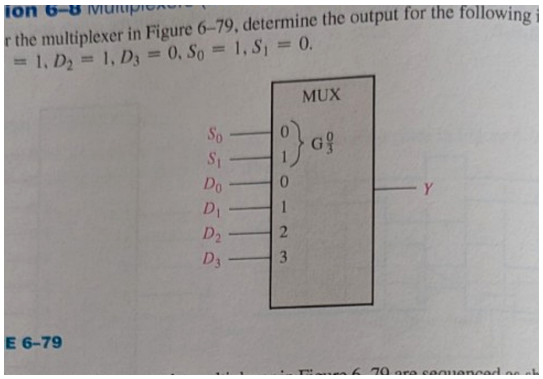


FIGURE 6–79

5. **[Section 6-8, Exercise 30]**
   The waveforms in Figure 6–81 are observed on the inputs of a 74HC151 8-input multiplexer. Sketch the Y output waveform.

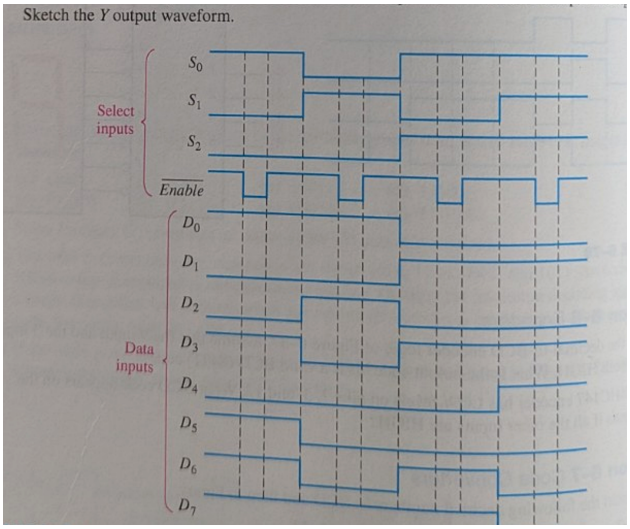


FIGURE 6–81

6. **[Special Design Problems, Exercise 46]**
Implement the logic function specified in Table 6–14 by using a 74HC151 data selector.

Inputs				Output
A_3	A_2	A_1	A_0	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

TABLE 6–14

2 Chapter 7

7. **[Section 7-1, Exercise 7]**
For a gated D latch, the waveforms shown in Figure 7–76 are observed on its inputs. Draw the timing diagram showing the output waveform you would expect to see at Q if the latch is initially RESET.

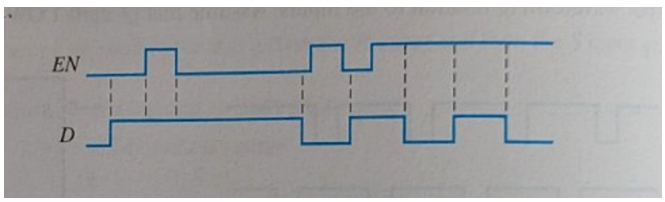


FIGURE 7–76

8. **[Section 7-2, Exercise 8]**
Two edge-triggered J-K flip-flops are shown in Figure 7–77. If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET.

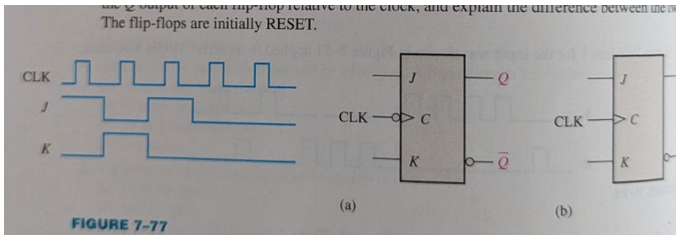


FIGURE 7–77

9. **[Section 7-2, Exercise 14]**
Determine the Q waveform relative to the clock if the signals shown in Figure 7–83 are applied to the inputs of the J-K flip-flop. Assume that Q is initially LOW.

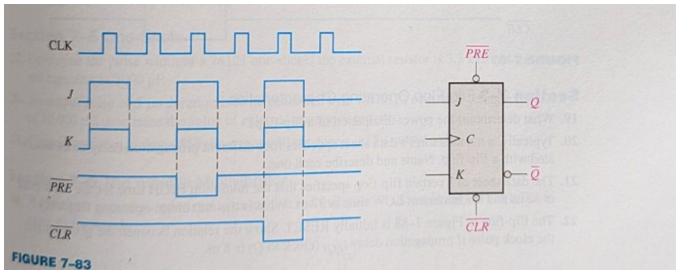


FIGURE 7–83

10. **[Section 7-7, Exercise 33]**
A 74HC00 quad NAND gate IC is used to construct a gated S-R latch on a protoboard in the lab as shown in Figure 7–93. The schematic in part (a) is used to connect the circuit in part (b). When you try to operate the latch, you find that the Q output stays HIGH no matter what the inputs are. Determine the problem.

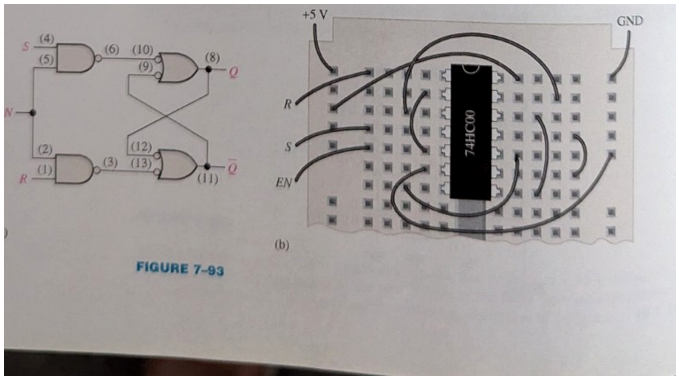


FIGURE 7–93

3 Chapter 8

11. **[Section 8-2, Exercise 14]**
The shift register in Figure 8–54(a) has $\overline{\text{SHIFT/LOAD}}$ and CLK inputs as shown in part (b). The serial data input (SER) is a 0. The parallel data inputs are $D_0 = 1$, $D_1 = 0$, $D_2 = 1$, and $D_3 = 0$ as shown. Develop the data-output waveform in relation to the inputs.

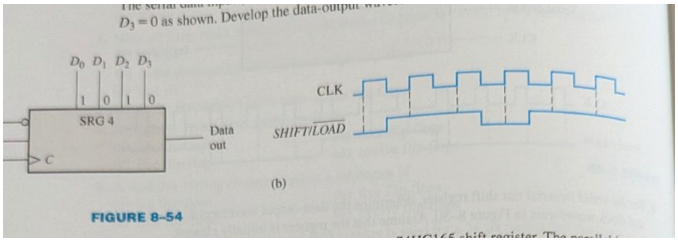


FIGURE 8–54

4 Chapter 9

12. **[Section 9-2, Exercise 3]**
For the ripple counter shown in Figure 9–65, show the complete timing diagram for eight clock pulses, showing the clock, Q_0, and Q_1 waveforms.

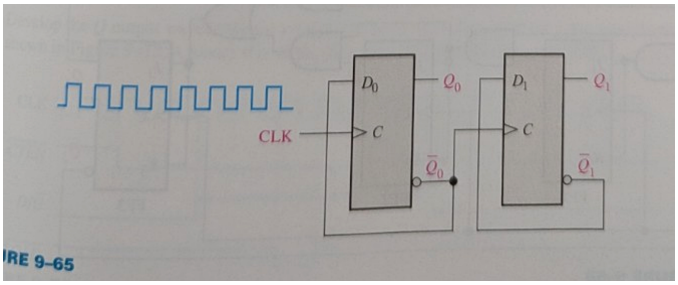


FIGURE 9–65